Report on Task 1

Title: Rice Variety Classification from Microscopic Images using CNN with Attention

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1. Introduction

Rice variety identification is an important task for quality control, seed purity verification, and precision agriculture. Manual identification by human experts is time-consuming, inconsistent, and dependent on individual experience. This project aims to automate the classification of processed rice varieties using microscopic images, by building a Convolutional Neural Network (CNN) enhanced with an attention mechanism (Track 3).

Before any model design, it is essential to understand the structure, quality, and characteristics of the dataset. This report presents a detailed Exploratory Data Analysis (EDA) of the chosen dataset, the PRBD (Processed Rice Bangladesh Dataset) covering class balance, visual similarity between classes, resolution consistency, colour/brightness behaviour, and image quality. The findings guide key decisions for the modeling stage in Task 2, including preprocessing steps, evaluation metric choice, and a leakage-safe data-splitting strategy.

2. Dataset Details

This project uses the PRBD (Processed Rice Bangladesh Dataset), obtained from Mendeley Data (data.mendeley.com/datasets/sfp9s96prh/1). The dataset consists of microscopic images of ten different processed rice varieties commonly found in Bangladesh - Aush, Beroi, BR-28, BR-29, Chinigura, Ghee Bhog, Katari Najir, Katari Siddho, Miniket, and Swarna - all collected from the local markets of Dhaka, Bangladesh. It is organized into two folders: Original_Images, containing 2000 high-resolution microscopic images (200 per class) in JPG format, and Augmented_images, containing 8000 pre-augmented images generated from the originals through transformations such as rotation, flipping, and scaling. For this project, only the Original_Images folder is used for exploratory data analysis and for building the train/validation/test split, since using the pre-augmented images for splitting could allow near-duplicate copies of the same source image to appear in both the training and test sets, causing data leakage and an artificially inflated accuracy. Any augmentation required for model training will instead be applied by us, only on the training split, during Task 2. The problem is framed as a multi-class image classification task, where the goal is to correctly identify the rice variety from a given microscopic grain image.

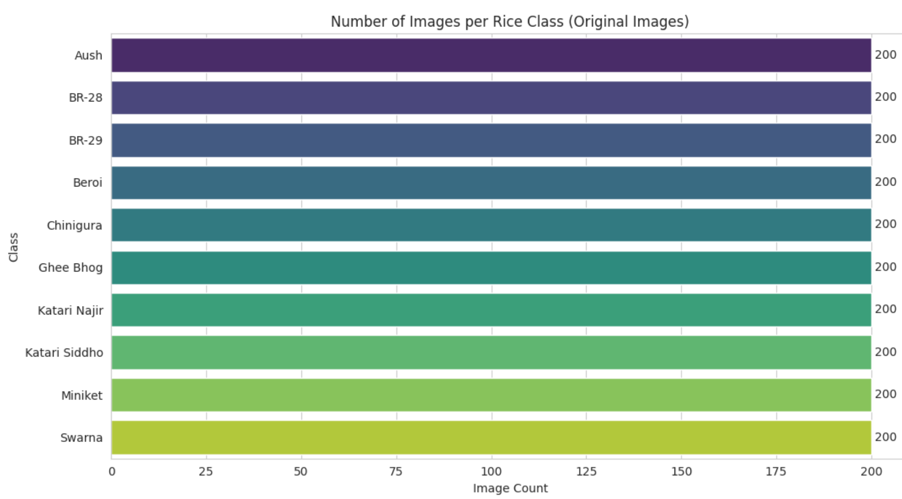
3. Exploratory Data Analysis

3.1 Data Loading & Summary

All 2000 image file paths were scanned and indexed across the 10 class subfolders, confirming exactly 2000 images in JPG format, matching the dataset's stated size and structure.

3.2 Class Balance

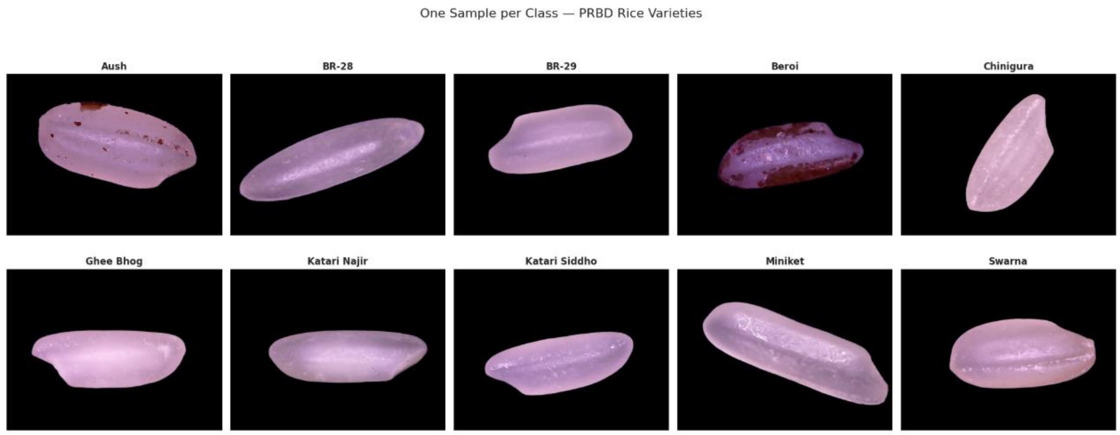
The number of images per class was counted and plotted. Every one of the 10 classes contains exactly 200 images, giving a majority-to-minority ratio of 1.00 : 1 — the dataset is perfectly balanced. This means plain accuracy is a fair evaluation metric here, though macro-F1 will still be reported in Task 2 for completeness.




```
Majority class: Aush (200 images)
Minority class: Aush (200 images)
Majority:Minority ratio = 1.00 : 1
```

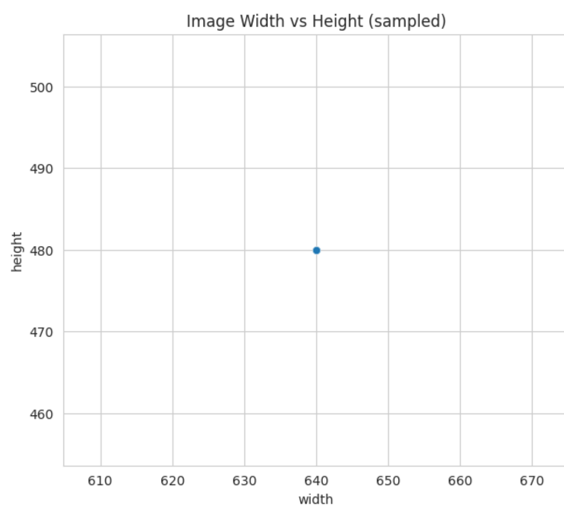
3.3 Labelled Sample Grid

One random sample image from each class was displayed for visual comparison. Most varieties share a similar elongated grain shape, so shape alone is not a strong distinguishing feature. Beroi, Katari Najir, and Katari Siddho appear visually closest to each other, making them likely confusion pairs, while Beroi and Aush show a distinct reddish/brown husk residue that could act as an unintended shortcut feature, to be checked later with Grad-CAM in Task 3.



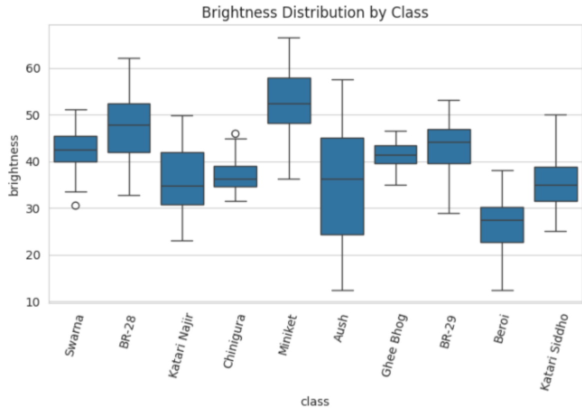
3.4 Size / Resolution Check

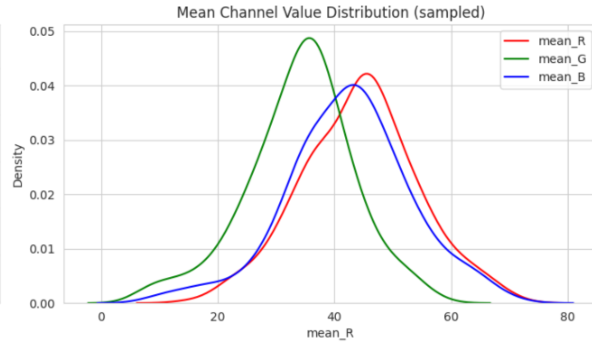
Image dimensions were measured on a sample of 500 images. All images share a single uniform resolution of 640 × 480 pixels with zero variation, so all images will be resized to 224 × 224 for CNN input in Task 2 without risk of distortion inconsistency.



3.5 Pixel / Colour Analysis

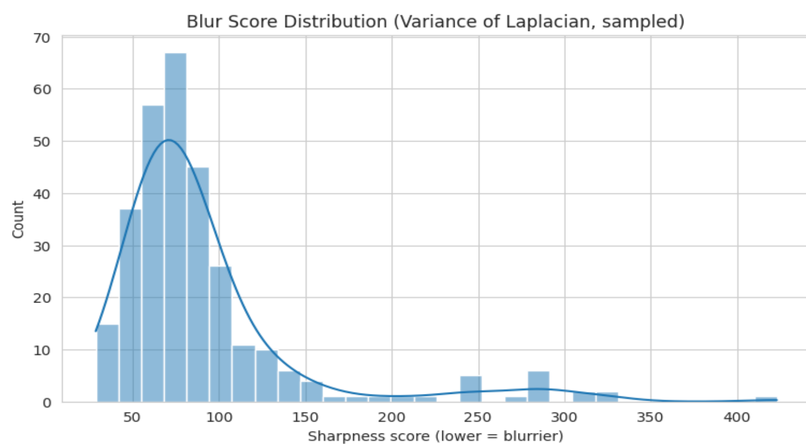
Mean RGB values and brightness were computed for a sample of 300 images. Brightness and colour distributions overlap heavily across all classes with no class standing out, confirming that colour/lighting is not a usable shortcut and the model will need to rely on finer texture cues, supporting the use of an attention mechanism.





3.6 Image Quality Check

Corrupt-file, blur, and duplicate checks were performed. No corrupt files (0/2000) and no exact duplicates were found. A few comparatively blurry images, mostly from Katari Najir, BR-29, and Swarna, were flagged for manual review but do not require removal.




```
Blurriest 5 images (lowest scores, candidates to inspect):
             class                      filename  blur_score
1323  Katari Najir  processed_S20250325_0151.jpg   28.557370
584          BR-29  processed_S20250323_0002.jpg   28.795241
464          BR-29  processed_S20250323_0045.jpg   31.474798
1301  Katari Najir  processed_S20250325_0075.jpg   34.014382
1884        Swarna  processed_S20250322_0007.jpg   34.690091
```

3.7 Interactive Plots

Interactive Plotly versions of the class-balance, resolution, and brightness charts were also generated for live inspection during presentation, available in the accompanying notebook.